Userflow Doc

This is the level you can hand to:

- designers → for Figma flows
- developers → for implementation
- product → for roadmap

🧠 SYSTEM CORE (MENTAL MODEL)

```
Super Admin → Institute → Owner → Staff (RBAC) → Operations → Student/Parent
```

👉 Everything revolves around this loop:

```
Student ↔ Batch ↔ Attendance ↔ Fees ↔ Payments ↔ Notifications ↔ Reports
```

If this loop breaks → product fails.

👑 1. SUPER ADMIN FLOW (SaaS CONTROL LAYER)

🎯 Purpose:

Control tenants, billing, access, and platform stability

🔁 MASTER FLOW

```
🔵 Login

→ 🟩 Dashboard
    - KPIs: Revenue, Active Institutes, Expiry Alerts

→ 🟩 Institutes Module
```

```
→ 🟨 Action Selection
    ├── Create Institute →
    │     🟩 Enter Institute + Owner Info
    │     → 🟩 Assign Plan (Trial/Paid)
    │     → 🟩 Generate Credentials
    │     → 🟩 Notify Owner
    │     → 🔴 End
    │
    ├── View Institute →
    │     🟩 Open Profile
    │     → 🟩 Tabs:
    │         - Users
    │         - Billing
    │         - Usage
    │
    │     → 🟨 Action Needed?
    │         ├── Suspend →
    │         │     🟩 Disable Login
    │         │     → 🔴 End
    │         │
    │         ├── Upgrade Plan →
    │         │     🟩 Change Limits
    │         │     → 🔴 End
    │         │
    │         ├── Impersonate →
    │         │     🟩 Login as Owner
    │         │     → 🔴 End
    │         │
    │         └── No Action → 🔴 End
    │
    └── Ignore → 🔴 End
```

⚠️ EDGE CASES (SUPER ADMIN)

Case	Type
Institute subscription expired but still active	🔴 Critical
Owner deleted but institute exists	🔴 Critical
Multiple logins via impersonation conflict	🟡 Normal
Data isolation failure between tenants	🔴 Critical
Plan downgrade causing data overflow	🔴 Critical

🏫 2. INSTITUTE OWNER FLOW (CONTROL + DELEGATION)

🎯 Purpose:

- Configure system
- Delegate operations
- Monitor everything

🔁 MASTER OWNER FLOW

```
🔵 Login

→ 🟩 Dashboard

→ 🟨 Select Intent
    ├── Setup System
    ├── Manage Operations
    ├── Delegate Work (Staff)
    ├── Monitor Reports
```

⚙️ A. INITIAL SETUP FLOW (FIRST TIME EXPERIENCE)

```
🟩 First Login

→ 🟩 Create Fee Plans
    → Monthly / Course / Installment

→ 🟩 Create Batches
    → Assign Timing
    → Assign Subject

→ 🟩 Add Staff (optional early)

→ 🟩 Add Students
    → Assign Batch
    → Auto Fee Mapping

→ 🟩 System Ready
```

⚠️ EDGE CASES (SETUP)

Case	Type
Student added without batch	🔴 Critical
Batch without fee plan	🔴 Critical
Teacher not assigned but batch active	🟡 Normal
Duplicate student (same phone)	🟡 Normal

👨‍💼 B. STAFF CREATION + PERMISSION FLOW (CORE ENGINE)

🎯 This is your most important system logic

```
🟩 Owner → Staff Module

→ 🟩 Add Staff

→ 🟩 Enter Details

→ 🟩 Select Role
    ├── Teacher
    ├── Accountant
    ├── Admin/Staff

→ 🟨 Customize Permissions?
    ├── Yes →
    │      🟩 Permission Matrix:
    │          - Students (View/Edit/Delete)
    │          - Fees (Collect/Edit)
    │          - Attendance (Mark/Edit)
    │          - Notifications (Send)
    │          - Reports (View)
    │
    │      → 🟩 Save Custom Role
    │      → 🔴 End
    │
    └── No →
           🟩 Apply Default Permissions
           → 🔴 End
```

🔁 PERMISSION EXECUTION FLOW

```
🟩 Staff Login

→ 🟩 Load Allowed Modules

→ 🟨 Access Request
    ├── Allowed →
    │      🟩 Execute Action
    │      → 🔴 End
    │
    └── Denied →
           🟩 Show Error
           → 🔴 End
```

⚠️ EDGE CASES (PERMISSIONS)

Case	Type
Staff edits fees without permission	🔴 Critical
Owner removes own permissions accidentally	🔴 Critical
Role updated but session not refreshed	🟡 Normal
Multiple roles assigned conflicting permissions	🟡 Normal

🔄 C. DAILY OPERATIONS FLOW

👩‍🎓 STUDENT FLOW

```
🟩 Open Students

→ 🟨 Action
    ├── Add →
    │      🟩 Enter Info
    │      → 🟩 Assign Batch
    │      → 🟩 Link Fee Plan
    │      → 🔴 End
    │
    ├── Edit →
    │      🟩 Update Details
    │      → 🔴 End
    │
    └── View →
           🟩 Profile + History
           → 🔴 End
```

⚠️ EDGE CASES (STUDENT)

Case	Type
Student in multiple batches (fee conflict)	🔴 Critical
Removing student but payments exist	🔴 Critical
Changing batch mid-month	🟡 Normal
Missing parent contact	🟡 Normal

💰 FEES FLOW (MOST CRITICAL MODULE)

```
🟩 Open Fees

→ 🟩 View Dues

→ 🟨 Select Student
    ├── Yes →
    │     🟩 Enter Amount
    │     → 🟨 Full Payment?
    │         ├── Yes → Close Due
    │         └── No → Update Remaining
    │
    │     → 🟩 Generate Receipt
    │     → 🔴 End
    │
    └── No → 🔴 End
```

⚠️ EDGE CASES (FEES)

Case	Type
Partial payment not tracked correctly	🔴 Critical
Duplicate payment entry	🔴 Critical
Payment without receipt	🔴 Critical
Fee plan changed after payment	🔴 Critical
Refund handling missing	🟡 Normal
Offline payment sync issues	🔴 Critical

📢 NOTIFICATION FLOW

```
🟩 Trigger

→ 🟨 Type
    ├── Auto →
    │      🟩 Fee Reminder / Absence Alert
    │
    ├── Manual →
    │      🟩 Staff Sends Message
```

⚠️ EDGE CASES (NOTIFICATIONS)

Case	Type
Duplicate notifications sent	🟡 Normal
Wrong parent number	🔴 Critical
WhatsApp API failure	🔴 Critical
Spam complaints	🟡 Normal

👨‍🏫 3. TEACHER FLOW (HIGH FREQUENCY)

```
🔵 Login

→ 🟩 My Batches

→ 🟨 Select Batch
    ├── Yes →
    │      🟩 Mark Attendance
    │      → 🟨 Already Marked?
    │          ├── Yes → Lock
    │          └── No → Submit
    │
    │      → 🔴 End
    │
    └── No → 🔴 End
```

⚠️ EDGE CASES (TEACHER)

Case	Type
Attendance marked twice	🔴 Critical
Wrong date marking	🔴 Critical
Teacher assigned to wrong batch	🟡 Normal
Offline attendance not syncing	🔴 Critical

👨‍💼 4. STAFF FLOW (EXECUTION LAYER)

```
🔵 Login

→ 🟩 Load Modules (based on permissions)

→ 🟨 Select Task
    ├── Fees
    ├── Students
    ├── Notifications
    ├── Reports

→ 🟩 Execute Task

→ 🔴 End
```

⚠️ EDGE CASES (STAFF)

Case	Type
Staff accessing restricted data	🔴 Critical
Staff deletes important data	🔴 Critical
Session hijacking	🔴 Critical
Misconfigured permissions	🔴 Critical

🎓 5. STUDENT / PARENT FLOW

```
🔵 Login

→ 🟩 Dashboard

→ 🟨 Select
    ├── Attendance
```

```
    ├── Fees
    ├── Notifications

→ 🟩 View Data

→ 🔴 End
```

⚠️ EDGE CASES (STUDENT)

Case	Type
Wrong student mapped to parent	🔴 Critical
Payment shown but not received	🔴 Critical
Attendance mismatch	🔴 Critical
Login OTP failure	🟡 Normal

🔁 SYSTEM AUTOMATION FLOW

```
🟩 Background Jobs

→ 🟨 Event
    ├── Fee Due →
    │      🟩 Trigger Reminder
    │
    ├── Absence →
    │      🟩 Notify Parent
    │
    └── Payment →
           🟩 Send Receipt
```

⚠️ EDGE CASES (SYSTEM)

Case	Type
Cron job failure	🔴 Critical
Notification delay	🔴 Critical
Data inconsistency between modules	🔴 Critical
Timezone issues	🟡 Normal

🔥 FINAL PRODUCT INSIGHT

This system is not just roles—it's:

👉 A Delegation Engine

```
Owner → Defines System
      → Assigns Roles
      → Controls Permissions

Staff → Executes Tasks

System → Automates + Tracks

Student → Receives Output
```

SYSTEM (Coaching ERP SaaS)
│
├── SUPER ADMIN (SaaS Owner)
│ │
│ ├── Authentication
│ │ └── Login → Dashboard
│ │
│ ├── Dashboard
│ │ ├── View KPIs (Revenue, Institutes, Expiry Alerts)
│ │ └── Navigate Modules
│ │
│ ├── Institute Management
│ │ │
│ │ ├── Create Institute
│ │ │ ├── Enter Institute Details
│ │ │ ├── Assign Plan (Trial/Paid)
│ │ │ └── Generate Owner Credentials
│ │ │
│ │ ├── View Institute
│ │ │ ├── Users
│ │ │ ├── Billing
│ │ │ └── Usage
│ │ │

```
│   │   ├── Actions
│   │   │   ├── Upgrade/Downgrade Plan
│   │   │   ├── Suspend/Activate Institute
│   │   │   └── Impersonate Owner
│   │   │
│   │   └── EDGE CASES
│   │       ├── 🔴 Data leakage across tenants
│   │       ├── 🔴 Expired plan but active access
│   │       └── 🟡 Concurrent impersonation conflict
│   │
│   └── System Control
│       └── Global Settings / Logs
│
├── INSTITUTE OWNER (TENANT ADMIN)
│   │
│   ├── Authentication
│   │   └── Login → Dashboard
│   │
│   ├── INITIAL SETUP
│   │   │
│   │   ├── Fee Plans
│   │   │   ├── Monthly
│   │   │   ├── Course
│   │   │   └── Installment
│   │   │
│   │   ├── Batch Setup
│   │   │   ├── Create Batch
│   │   │   ├── Assign Teacher
│   │   │   └── Assign Fee Plan
│   │   │
│   │   ├── Add Students
│   │   │   ├── Enter Details
│   │   │   ├── Assign Batch
│   │   │   └── Auto Fee Mapping
│   │   │
│   │   └── EDGE CASES
│   │       ├── 🔴 Student without batch
│   │       ├── 🔴 Batch without fee plan
```

```
│   │       └── 🟡 Duplicate student
│   │
│   ├── STAFF MANAGEMENT (DELEGATION ENGINE)
│   │   │
│   │   ├── Add Staff
│   │   │   ├── Enter Details
│   │   │   ├── Assign Role
│   │   │   │   ├── Teacher
│   │   │   │   ├── Accountant
│   │   │   │   └── Admin/Staff
│   │   │   │
│   │   │   └── Customize Permissions
│   │   │       ├── Students (View/Edit/Delete)
│   │   │       ├── Fees (Collect/Edit)
│   │   │       ├── Attendance (Mark/Edit)
│   │   │       ├── Notifications (Send)
│   │   │       └── Reports (View)
│   │   │
│   │   ├── Edit Staff
│   │   │   └── Update Permissions
│   │   │
│   │   ├── Remove Staff
│   │   │   └── Deactivate Account
│   │   │
│   │   └── EDGE CASES
│   │       ├── 🔴 Owner removes own access
│   │       ├── 🔴 Unauthorized permission escalation
│   │       └── 🟡 Role update not reflected instantly
│   │
│   ├── DAILY OPERATIONS
│   │   │
│   │   ├── Student Management
│   │   │   ├── Add Student
│   │   │   ├── Edit Student
│   │   │   ├── Assign / Change Batch
│   │   │   └── View Profile (Fees + Attendance)
│   │   │
│   │   │   └── EDGE CASES
```

```
│   │   │       ├── 🔴 Student in multiple batches (fee conflict)
│   │   │       ├── 🔴 Delete student with payments
│   │   │       └── 🟡 Batch switch mid-cycle
│   │   │
│   │   ├── Batch Management
│   │   │   ├── Create Batch
│   │   │   ├── Assign Teacher
│   │   │   ├── Assign Students
│   │   │   └── Modify Schedule
│   │   │
│   │   │   └── EDGE CASES
│   │   │       ├── 🔴 Batch active without teacher
│   │   │       └── 🟡 Time conflict between batches
│   │   │
│   │   ├── Fees Management (MOST CRITICAL)
│   │   │   ├── View Dues
│   │   │   ├── Collect Payment
│   │   │   │   ├── Enter Amount
│   │   │   │   ├── Select Mode (Cash/UPI/Bank)
│   │   │   │   └── Generate Receipt
│   │   │   │
│   │   │   ├── Handle Partial Payments
│   │   │   └── Track Defaulters
│   │   │
│   │   │   └── EDGE CASES
│   │   │       ├── 🔴 Duplicate payment entry
│   │   │       ├── 🔴 Payment without receipt
│   │   │       ├── 🔴 Fee plan changed mid-cycle
│   │   │       ├── 🔴 Data mismatch (paid vs due)
│   │   │       └── 🟡 Refund handling
│   │   │
│   │   ├── Notifications
│   │   │   ├── Auto Triggers
│   │   │   │   ├── Fee Due Reminder
│   │   │   │   └── Absence Alert
│   │   │   │
│   │   │   └── Manual Messaging
│   │   │
```

```
│   │   │   └── EDGE CASES
│   │   │       ├── 🔴 Wrong recipient
│   │   │       ├── 🔴 API failure (WhatsApp/SMS)
│   │   │       └── 🟡 Duplicate messages
│   │   │
│   │   └── Reports & Monitoring
│   │       ├── Revenue
│   │       ├── Attendance
│   │       └── Defaulters
│   │
│   └── SYSTEM CONTROL LOOP
│       └── Monitor → Adjust → Delegate → Repeat
│
├── TEACHER
│   │
│   ├── Login
│   │   └── View Assigned Batches
│   │
│   ├── Attendance Flow
│   │   ├── Select Batch
│   │   ├── Mark Attendance
│   │   ├── Submit
│   │   └── View History
│   │
│   └── EDGE CASES
│       ├── 🔴 Attendance marked twice
│       ├── 🔴 Wrong date entry
│       ├── 🔴 Offline sync failure
│       └── 🟡 Wrong batch assignment
│
├── STAFF (ROLE-BASED EXECUTION)
│   │
│   ├── Login
│   │   └── Load Allowed Modules (RBAC)
│   │
│   ├── Modules (based on permission)
│   │   ├── Fees
│   │   ├── Students
```

```
│   │   ├── Notifications
│   │   └── Reports
│   │
│   ├── Action Execution
│   │   └── Perform Allowed Tasks Only
│   │
│   └── EDGE CASES
│       ├── 🔴 Unauthorized access
│       ├── 🔴 Data deletion without audit
│       ├── 🔴 Session hijack
│       └── 🟡 UI mismatch after permission change
│
├── STUDENT / PARENT
│   │
│   ├── Login (OTP-based)
│   │
│   ├── Dashboard
│   │   ├── Attendance %
│   │   └── Fees Status
│   │
│   ├── Modules
│   │   ├── Attendance History
│   │   ├── Fees (Paid / Due)
│   │   └── Notifications
│   │
│   └── EDGE CASES
│       ├── 🔴 Wrong student linked to parent
│       ├── 🔴 Payment mismatch
│       ├── 🔴 Attendance mismatch
│       └── 🟡 OTP failure
│
└── SYSTEM AUTOMATION (BACKGROUND)
    │
    ├── Event Triggers
    │   ├── Fee Due → Send Reminder
    │   ├── Absence → Notify Parent
    │   └── Payment → Send Receipt
    │
```

└── EDGE CASES
├── 🔴 Cron job failure
├── 🔴 Notification delay
├── 🔴 Data inconsistency
└── 🟡 Timezone issues